



Instructor: Date:

Learner: Regional Branch:

Program: Center:

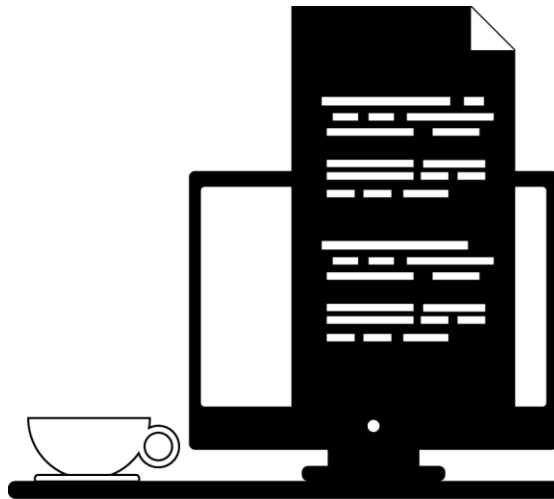
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Let's Start!

Getting Familiar With JavaScript



Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts. Describe applications for an operating system for personal computers in a range of work contexts.

Can Do Statements:

- I can say what Javascript is.
- I can talk about client side Javascript.
- I can discuss advantages and limitations of Javascript.

Ask your partner:

- How do you think webpages and software programs are created?
- How do you imagine the first computer programs looked like?





Let's Explore!

Camille gives Frank a hand* to understand JavaScript.

*To "give a hand" means to help someone.



Instructions: Read the text below.

Frank: I know you already took the programming course. Would you help me understand some things about JavaScript?

Camille: Sure! JavaScript is a dynamic programming language for browsers, servers, games and even robot applications that direct the behavior of the different components of a website.

Frank: But why is it so important if there are more programming languages already?

Camille: Because it improves the user's experience. If most people weren't so visual, it wouldn't be as popular. Moreover, there are some important features you must take into account if you haven't used JavaScript before.

First, JavaScript is a lightweight program which means that it doesn't need much memory to run and it's easy to implement. Second, it allows the client-side script that means it runs on the user's computer or browser. Third, it is object-oriented which focuses on objects presented in a web page, rather than the logic necessary for their manipulation. And fourth, it has network-centric characteristics which mean that it interconnects people, services, information and devices aiming to make the most out of resources.

Frank: Could we further discuss what Client-side JavaScript is?

Camille: Sure, this is the most common form of language. If JavaScript didn't allow client-side scripts, it wouldn't be as interactive as it permits the user to have more control over the program.

Frank: Wow, it sounds like a very complete application.

Camille: But it's not. It also has its limitations that need to be considered. For instance, reading or writing files is not allowed for security reasons. If you are planning to use JavaScript for networking applications, you won't be able to do so as it's not available, and it doesn't count with multiprocessor capabilities either.

Frank: True, but you can't deny the advantages that it has. Less server interaction lessens the load on user's server. Immediate feedback to visitors reduces time. And along with the increased interactivity, there is a richer interface.

Camille: You're right. Those are things we can't overlook.





Let's Practice!

Camille and Frank are talking about JavaScript.

Instructions: Read the sentences. Underline all the correct answers. Read the correct answers aloud. Check your pronunciation with your instructor.

1. Javascript, which is a lightweight program, _____.
 - a. was written to control information.
 - b. was designed to create network-centric applications.
 - c. was created to make things more complex.
 - d. was programmed to make things more dynamic.
2. JavaScript is a dynamic programming language for browsers, servers, games and even robot applications _____.
 - a. whom controls the way different components of a website work.
 - b. which works only in simple web levels.
 - c. that direct behavior of the different components of a website.
 - d. that control the way different components of websites behave.
3. JavaScript is client-side which means _____.
 - a. it runs in the user's computer or browser.
 - b. it only works in company servers.
 - c. it allows users to have more interaction with the program.
 - d. it is difficult to work with at home.



Let's Practice!

Frank and Camille discuss the pros and cons of JavaScript.

Instructions: Underline the correct answer.

1. JavaScript is client-side _____.
(HINT: It can work in the client's computers without problem.)
 - a. who is always interested in computers.
 - b. which means it runs in the user's computer or browser.
 - c. that is a common way to say it is complex.
 - d. which means it only works for professionals.





Let's Practice!

2. JavaScript is object-oriented _____
(HINT: The objects are the most important elements.)
 - a. which focuses on procedures or step-by-step instructions.
 - b. which means it focuses on the development of code only.
 - c. which means it focuses on objects present in a webpage.
 - d. which focuses on instructions and procedures.
3. JavaScript has its limitations that need to be taken into account, _____
 - a. which means it is not perfect.
 - b. which means it has nothing to be corrected.
 - c. which means it will disappear soon.
 - d. which means it won't work in the future.
4. JavaScript doesn't count with multiprocessor capabilities _____
(HINT: Some people consider this as a disadvantage.)
 - a. which means apps work faster.
 - b. which is a very nice feature.
 - c. which is an advantage.
 - d. which is an important limitation.
5. JavaScript has less server interaction _____
(HINT: As a result, it reduces the load on the server.)
 - a. that means it lessens the load on the user's server.
 - b. that means it increases the load on the user's server.
 - c. that means client-side scripts are ruled out.
 - d. that means that a more powerful server is needed for carrying out processes.



Extension Activity

Make a presentation about other characteristics JavaScript might have.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- a) Do some more research on JavaScript and identify other characteristics different from the ones mentioned before.
- b) Select the aspects on which you want to focus for your presentation and make a poster or a slide show explaining each aspect.
- c) Share your findings with your classmates.

Self-Assessment

	Yes	No
1. I do some further research on JavaScript.	☐	☐
2. I organize and select the relevant information about JavaScript.	☐	☐
3. I present my findings to the class.	☐	☐



Target Words

- **behavior** (noun): the way someone acts or something works
e.g. JavaScript is a dynamic programming language for browsers, servers, games and even robot applications that directs the behavior of the different components of a website.
- **lightweight program** (adjective): a program that runs easily on a computer
e.g. JavaScript is lightweight which means that it doesn't need much memory to run and it's easy to implement.
- **object** (noun): in webpage design, it refers to the different elements present in a webpage or application
e.g. It is object-oriented which focuses on objects presented in a web page, rather than the logic necessary for their manipulation.
- **aim** (verb): to pursue an objective or a target
e.g. It has network-centric characteristics which means that it interconnects people, services, information and devices aiming to make the most out of resources.
- **lessen** (verb): to reduce or decrease the amount of something
e.g. Less server interaction lessens the load on user's server.
- **overlook** (verb): to omit or ignore something
e.g. Those are things we can't overlook.



Learning Points

Conditional sentences with if clauses-negative form

A conditional sentence states something that happens, will happen, could have happened, or could happen depending on a condition.

Conditional statements are called like this because they talk about situations that need a condition to be true or not. There are different types of **conditional sentences** that show the likelihood of a given result to take place or that speculate about what could have resulted from situations that didn't occur in the past or from unreal or imaginary events. Every type of conditional sentence has its own grammar structure.

The usual **conditional sentence** is composed of two clauses: the "**if clause**" that explains the condition, and the "**main clause**" that presents the result or consequence.

If you want to make a **negative conditional statement**, you can make either the "if clause" or the "result clause" negative, however, you have to pay attention to what you want to express. Let's take a look at some examples:





Learning Points

If you **don't** have money, you **can't** go.
If you **don't** have money, you can go.
If you have money, you **can't** go.

If you observe the examples carefully, you'll notice that the meaning of what is being uttered changes significantly with each sentence so, don't forget to bear in mind what you want to communicate when deciding which of the clauses is going to be negative.

Examples

- *If people weren't so visual, it wouldn't be as important.*
- *If JavaScript didn't allow client-side scripts, it wouldn't be as interactive as it permits the user to have more control over the program.*
- *If you want to use it for networking applications, you won't be able to do it because it's not available.*

Where is Javascript today?

There are different updates and versions for JavaScript.

There are some updates and versions that have gradually improved gradually JavaScript. However, it's been a long time since it was updated or improved. As a result, the latest update is the ECMAScript where JavaScript adapts itself to fulfill the ECMA International standards.

Even though changes are not notorious, it actually adds useful and more advanced features that have been requested by JavaScript users.